

Litesh

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Education

Indian Institute of Technology Madras

Bachelor of Science in Data Science & Applications

2024 – 2028

CGPA: **9.05 / 10**

Skills

Programming Languages: Python, JavaScript, Java

Web & Backend: Flask, React, Vue.js, Node.js, Express, HTML, CSS, Bootstrap

Data & ML: NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Regression & Classification Modeling, Cross-Validation, Hyperparameter Optimization

Databases: MySQL, MongoDB

Tools: Git, GitHub, Linux, Redis, Celery, REST APIs

Projects

Sanjeevani HMS — Hospital Management System | GitHub

- Built a production-grade full-stack Hospital Management System using Vue 3 (Pinia, Vue Router), Flask-RESTful, and SQLite, supporting three distinct roles: patient, doctor, and admin.
- Designed a normalized relational schema across 8+ models (Users, Doctors, Patients, Appointments, Treatments, Availability) and built 30+ REST API endpoints across 5 blueprints with Redis caching on read-heavy routes.
- Integrated Celery and Redis for async task processing: PDF bill generation, doctor profile exports, CSV exports, and daily appointment email reminders via automated Celery Beat scheduling.
- Implemented token-based authentication with Flask-Security, role-based access control with Vue Router navigation guards, and a profile completion gate enforced on every route navigation.

SketchQL — AI-powered Database Design SaaS | Live Demo | GitHub

- Designed and deployed a full-stack SaaS application using the MERN stack (MongoDB, Express, React, Node.js) for visual database schema modeling and AI-assisted SQL/Mongoose code generation.
- Integrated React Flow for interactive schema visualization and Gemini API for natural language-based schema generation.
- Implemented JWT authentication, REST APIs, and deployed to Vercel with a live production environment.

GaddiPark — Vehicle Parking Management System | GitHub

- Developed a multi-user parking management system using Flask, Jinja2 templates, and a relational SQL database with separate admin and user dashboards.
- Designed REST APIs for real-time spot allocation, booking management, wallet top-ups, and payment workflows, with role-based access control enforced throughout.

Comment Category Prediction — Multi-modal NLP Classification | GitHub

- Built a multi-modal classification pipeline combining text, numerical, and categorical features to predict comment categories (4 classes) optimized for Macro F1-score.
- Applied dual TF-IDF vectorization via FeatureUnion: word-level unigrams/bigrams (50k features) and character-level n-grams (100k features) to capture both semantics and structural patterns like slang and misspellings.
- Handled severe class imbalance (~58% majority, ~2% minority class) using custom per-class weights in SGDClassifier with ElasticNet regularization, tuned via RandomizedSearchCV.
- Built a unified ColumnTransformer pipeline integrating text, log-transformed engagement features, and one-hot encoded identity/categorical signals.

Movie Review Sentiment Analysis — Multi-class NLP | GitHub

- Implemented a multi-class sentiment classification pipeline (negative / neutral / positive) on raw review text using TF-IDF vectorization.
- Trained and compared Multinomial Naive Bayes, Logistic Regression, and SVM classifiers; achieved **0.6347 accuracy** on the Kaggle leaderboard.

Certifications & Training

Mathematical Foundations for Machine Learning — IIT Madras BS Program | Aug 2025

Workshop · 5th Aug – 15th Aug 2025 · Coordinated by Prof. Andrew Thangaraj